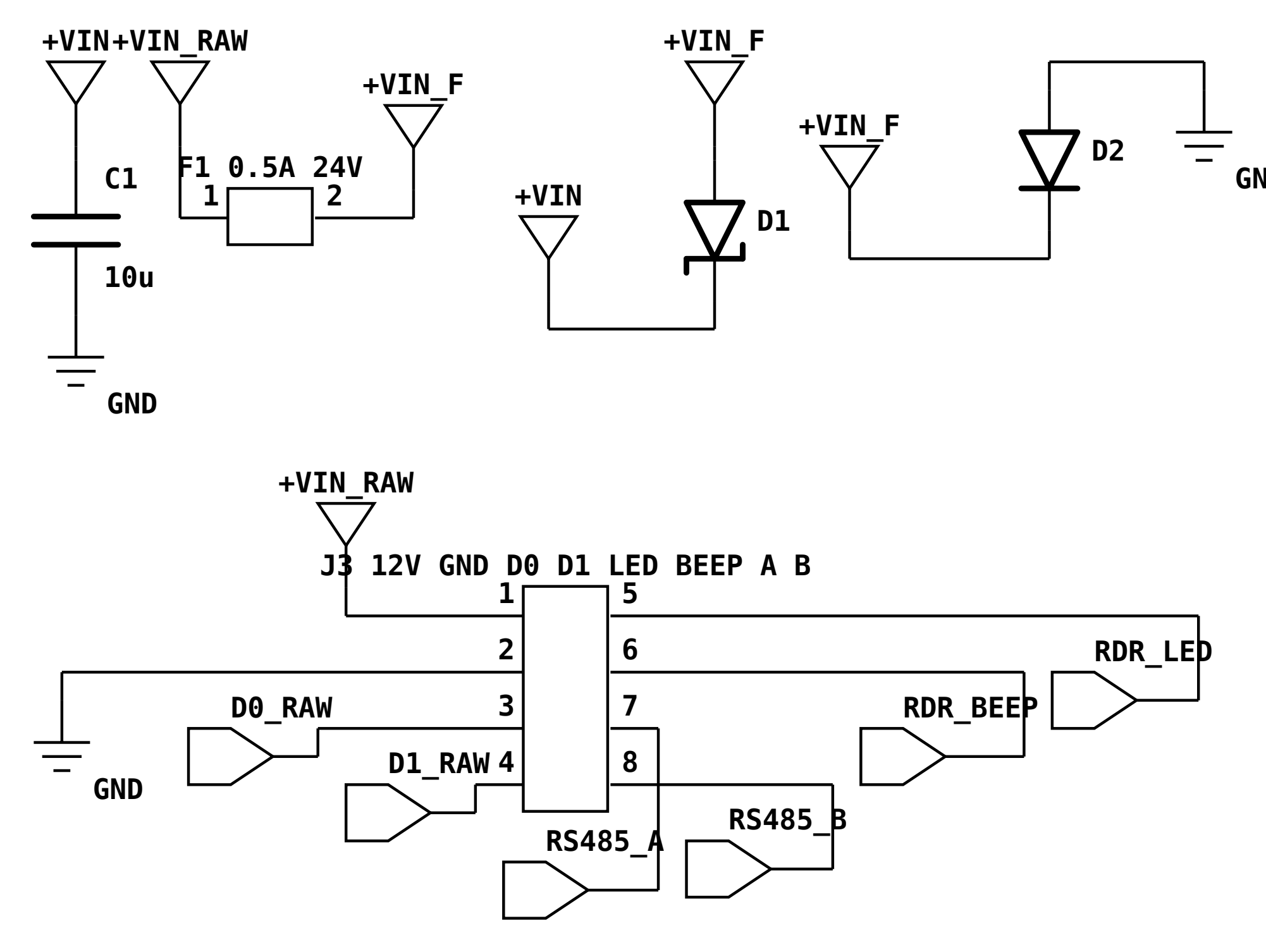
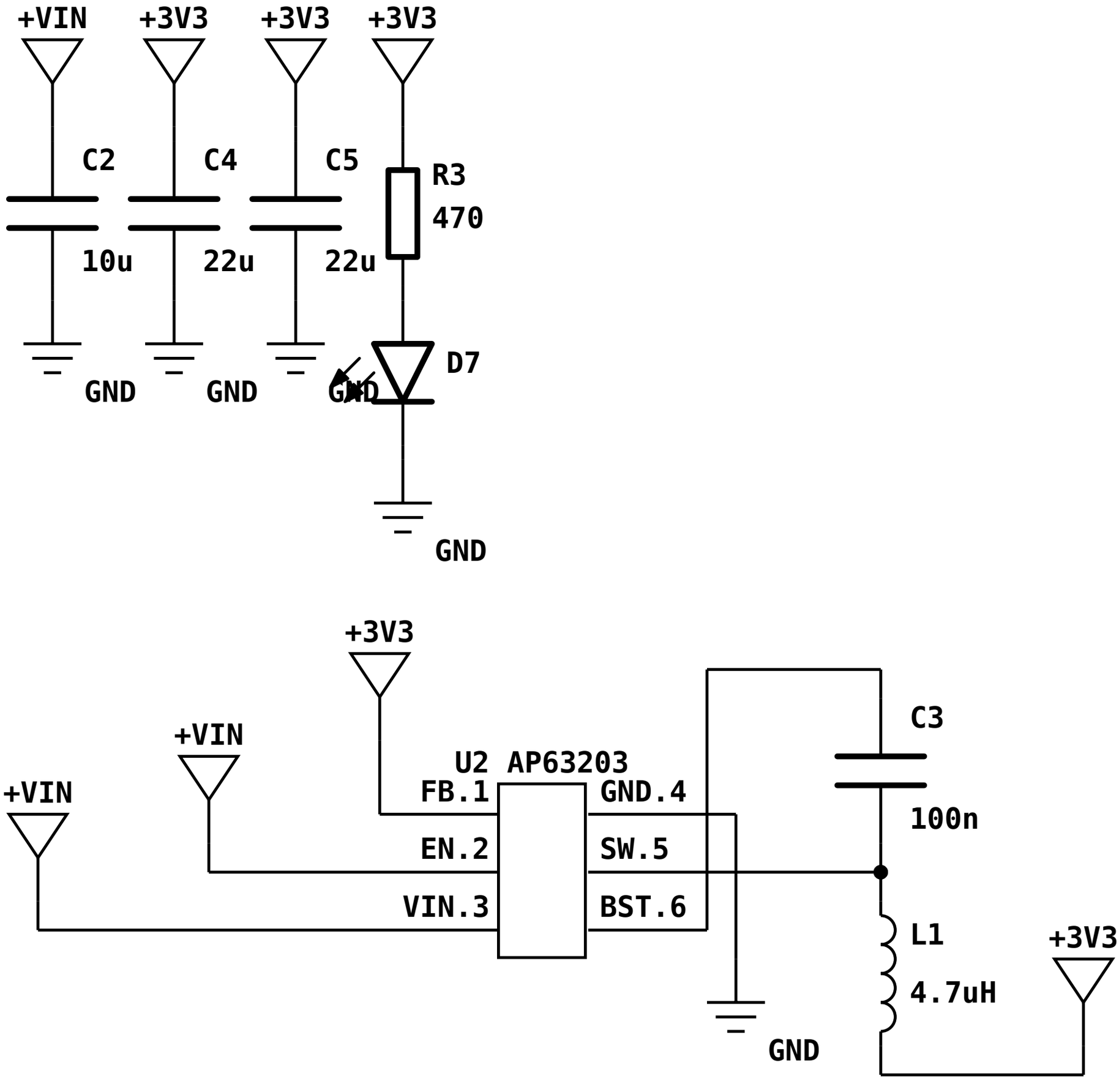
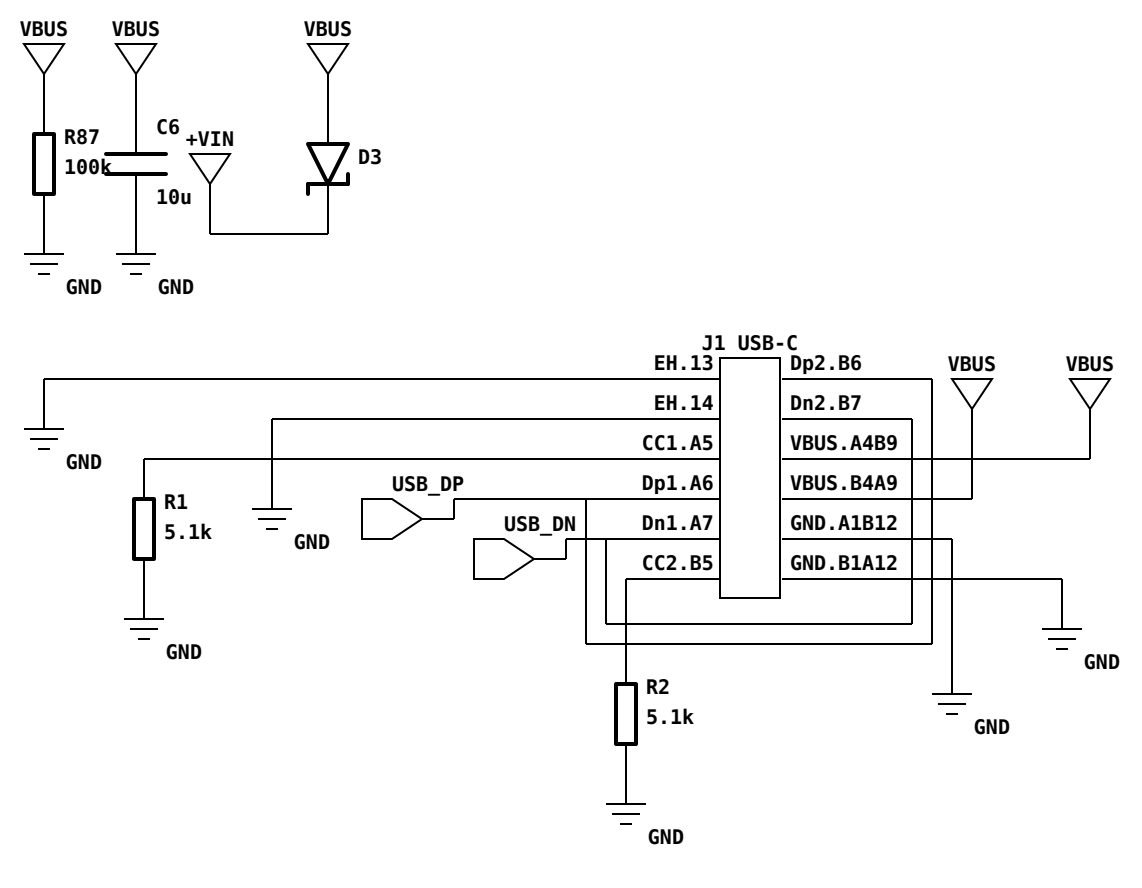
Wiegand bridge LITE — revision 1 schematic

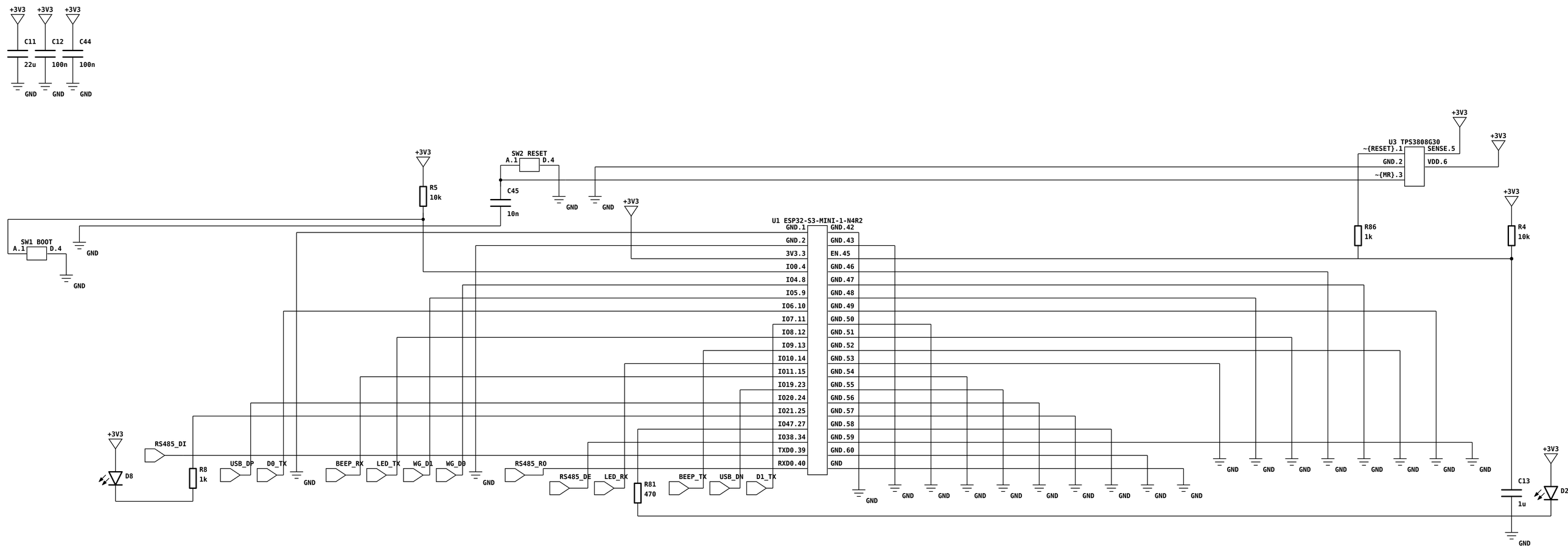
Generated from netlist.py (netlistsvg). Supply nets are flags; nets shared between blocks are named ports. I

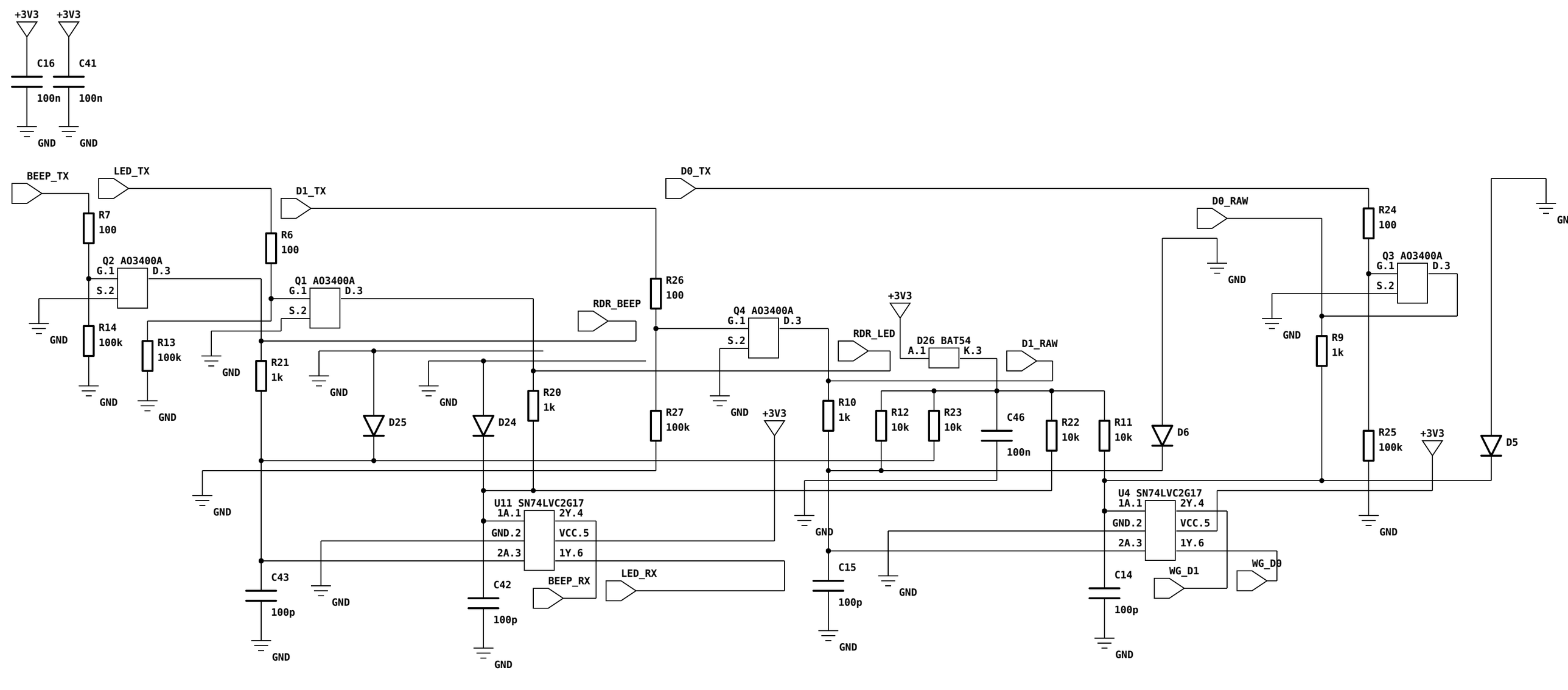
1. Block 1: terminal block and 8-16 V input
2. Block 2: buck to 3.3 V (AP63203, datasheet figure 21)
3. Block 3: USB-C (programming, and bench power through D3)
4. Block 4: MCU, buttons, status LEDs
5. Block 5: reader lines D0, D1, LED, BEEP -- bidirectional cells
6. Block 6: RS-485 / OSDP on UART0 (rev 10 block 14, less the bias network)
7. Block 7: surge/ESD on every field wire (SM712: +12 / -7 V standoff)

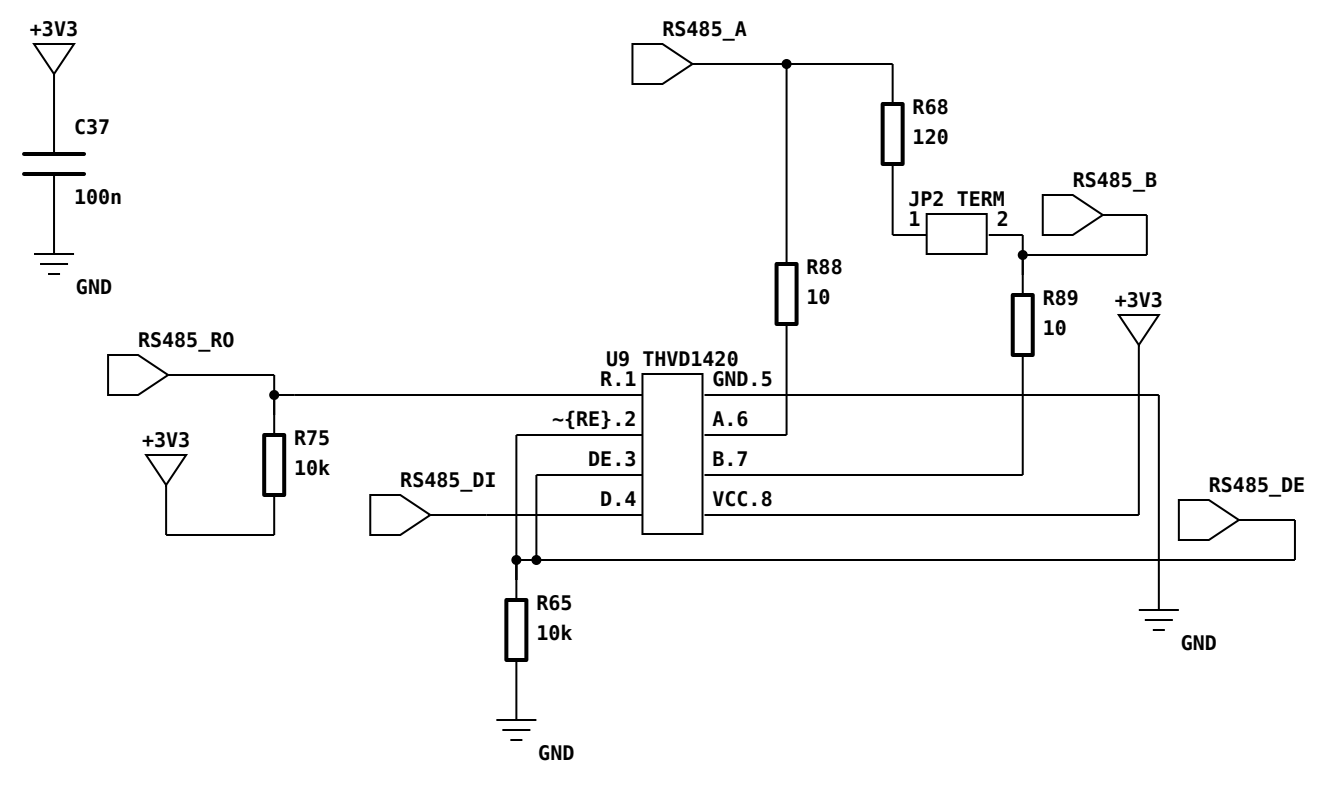

+VIN
+VIN_RAW
+VIN_F
C1
F1 0.5A 24V
1
2
10u
GND
+VIN_F
+VIN
D1
+VIN_F
D2
GN
+VIN_RAW
J3 12V GND D0 D1 LED BEEP A B
1
2
3
4
5
6
7
8
RDR_LED
D0_RAW
RDR_BEEP
GND
D1_RAW
RS485_B
RS485_A

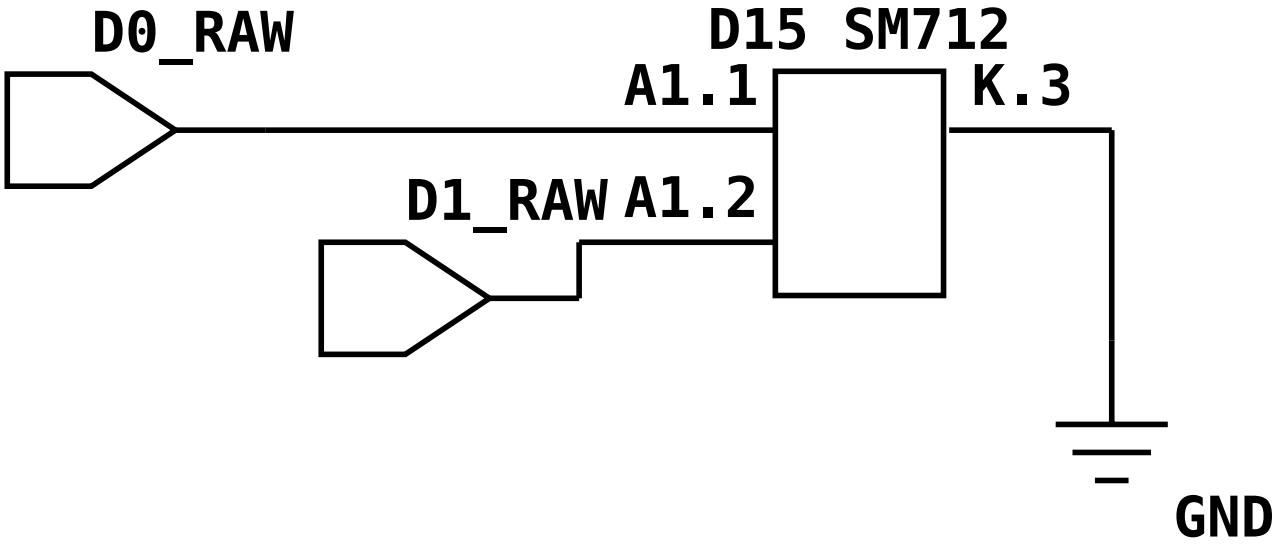

+VIN
+3V3
+3V3
+3V3
C2
C4
C5
R3
470
10u
22u
22u
GND
GND
GND
D7
GND
+3V3
+VIN
+VIN
U2 AP63203
FB.1
EN.2
VIN.3
GND.4
SW.5
BST.6
C3
100n
L1
4.7uH
+3V3
GND

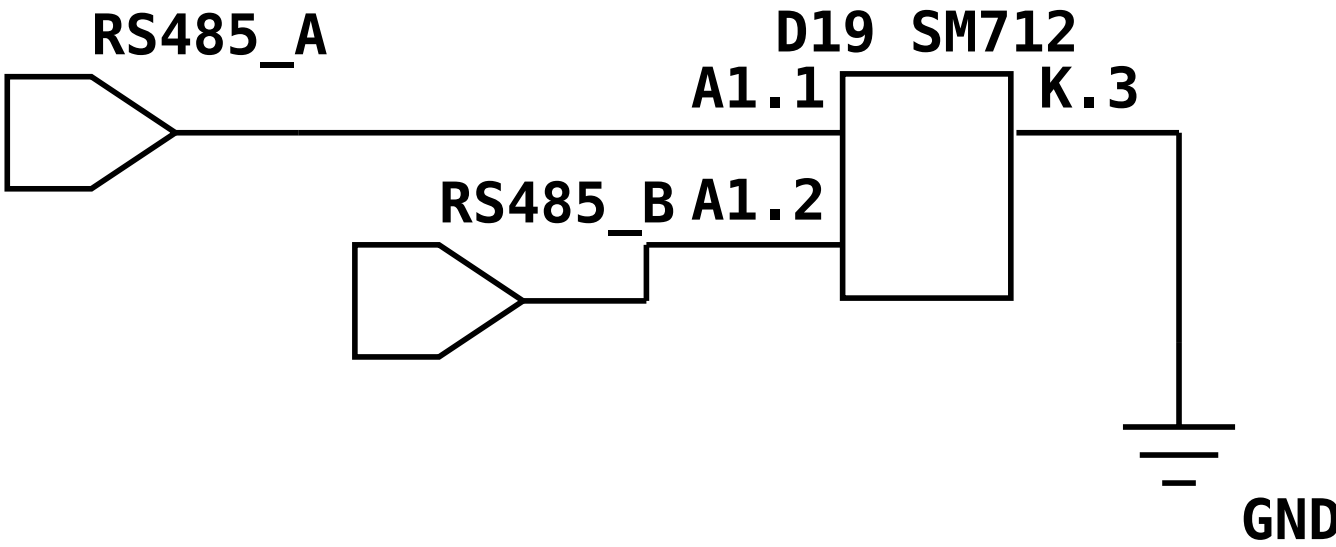

VBUS
VBUS
VBUS
R87
100k
C6
10u
+VIN
D3
GND
GND
J1 USB-C
EH.13
EH.14
CC1.A5
Dp1.A6
Dn1.A7
CC2.B5
Dp2.B6
Dn2.B7
VBUS.A4B9
VBUS.B4A9
GND.A1B12
GND.B1A12
VBUS
VBUS
GND
R1
5.1k
GND
USB_DP
USB_DN
GND
GND
R2
5.1k
GND
GND

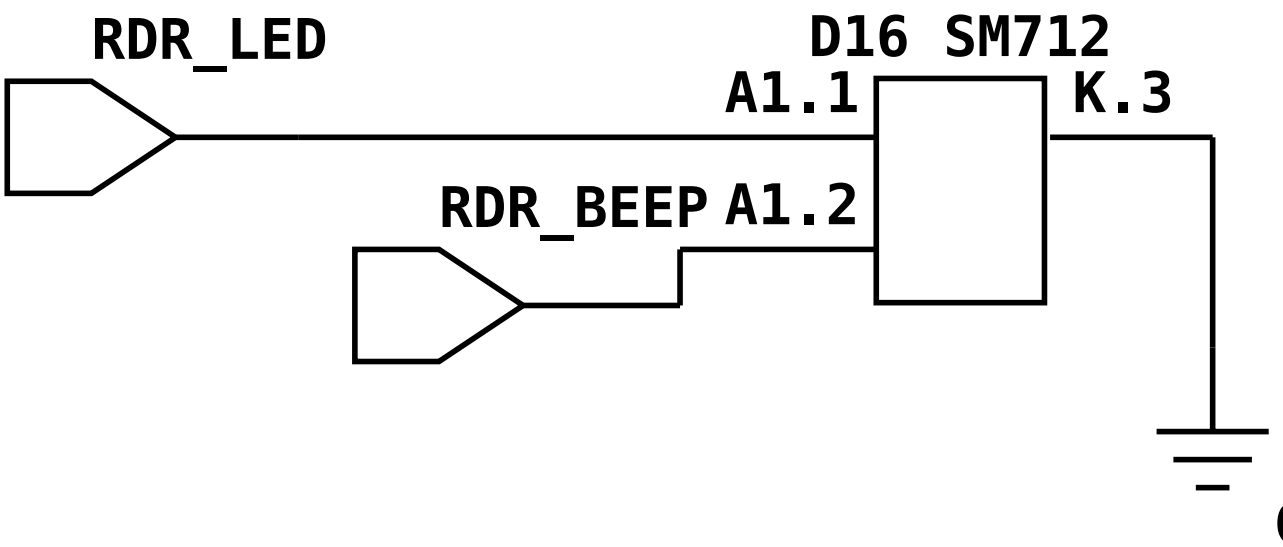

+3V3
C11
22u
GND
+3V3
C12
100n
GND
+3V3
C44
100n
GND
+3V3
R5
10k
SW2 RESET
A.1
D.4
C45
10n
GND
GND
+3V3
SW1 BOOT
A.1
D.4
GND
GND
U1 ESP32-S3-MINI-1-N4R2
GND.1
GND.2
3V3.3
IO0.4
IO4.8
IO5.9
IO6.10
IO7.11
IO8.12
IO9.13
IO10.14
IO11.15
IO19.23
IO20.24
IO21.25
IO47.27
IO38.34
TXD0.39
RXD0.40
GND.42
GND.43
EN.45
GND.46
GND.47
GND.48
GND.49
GND.50
GND.51
GND.52
GND.53
GND.54
GND.55
GND.56
GND.57
GND.58
GND.59
GND.60
GND
+3V3
U3 TPS3808G30
~{RESET}.1
GND.2
~{MR}.3
SENSE.5
VDD.6
+3V3
+3V3
R86
1k
R4
10k
+3V3
RS485_DI
R8
1k
D8
USB_DP
D0_TX
BEEP_RX
LED_TX
WG_D1
WG_D0
RS485_RO
RS485_DE
LED_RX
R81
470
BEEP_TX
USB_DN
D1_TX
C13
1u
D2
GND


+3V3
+3V3
C16
C41
100n
100n
GND
GND
BEEP_TX
LED_TX
D1_TX
D0_TX
D0_RAW
R7
100
R6
100
R26
100
R24
100
Q2 AO3400A
G.1
D.3
S.2
Q1 AO3400A
G.1
D.3
S.2
RDR_BEEP
Q4 AO3400A
G.1
D.3
S.2
+3V3
RDR_LED
D26 BAT54
A.1
K.3
D1_RAW
Q3 AO3400A
G.1
D.3
S.2
GND
R14
100k
R13
100k
R21
1k
R20
1k
R27
100k
R10
1k
R12
10k
R23
10k
C46
100n
R22
10k
R11
10k
R9
1k
R25
100k
D25
D24
D6
D5
U11 SN74LVC2G17
1A.1
2Y.4
GND.2
VCC.5
2A.3
1Y.6
U4 SN74LVC2G17
1A.1
2Y.4
GND.2
VCC.5
2A.3
1Y.6
C43
100p
C42
100p
C15
100p
C14
100p
BEEP_RX
LED_RX
WG_D1
WG_D0


+3V3
C37
100n
GND
RS485_A
R68
120
RS485_B
JP2 TERM
1
2
R88
10
R89
10
+3V3
RS485_RO
U9 THVD1420
R.1
GND.5
~{RE}.2
A.6
DE.3
B.7
D.4
VCC.8
+3V3
R75
10k
RS485_DI
RS485_DE
R65
10k
GND
GND


D0_RAW
D15 SM712
A1.1
K.3
D1_RAW
A1.2
GND


RS485_A
D19 SM712
A1.1
K.3
RS485_B
A1.2
GND


RDR_LED
D16 SM712
A1.1
K.3
RDR_BEEP
A1.2
GN